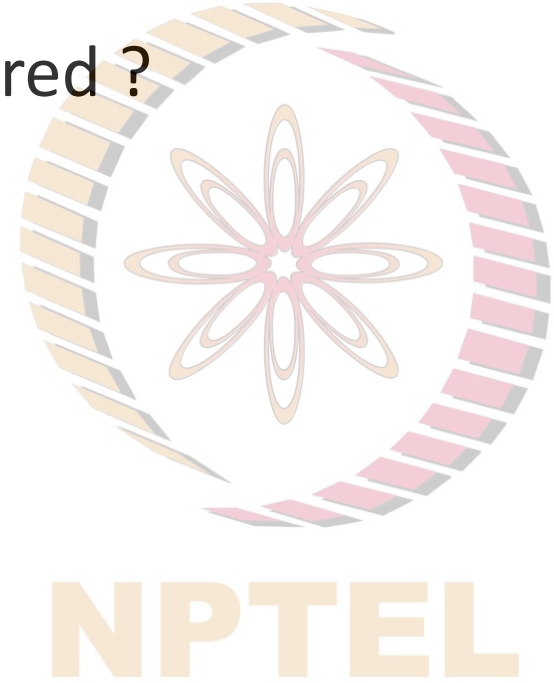
Packet Scheduling

- Why scheduling is required ?

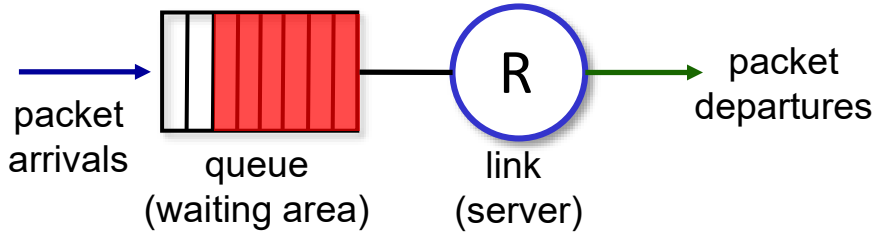


Packet Scheduling: FCFS

- packet scheduling: deciding which packet to send next on link
  - first come, first served
  - priority
  - round robin
  - weighted fair queueing

- FCFS: packets transmitted in order of arrival to output port
  - also known as: First-in-first-out (FIFO)

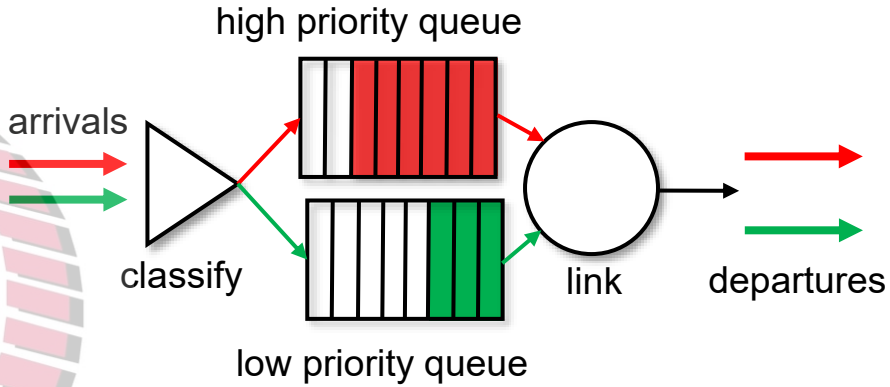
Abstraction: queue

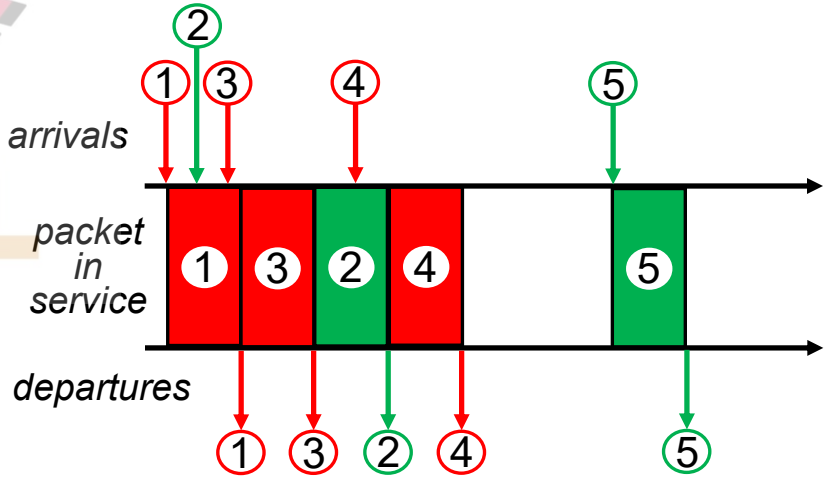


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Scheduling policies: priority

- *Priority scheduling:*
  - arriving traffic classified, queued by class
    - any header fields can be used for classification
  - send packet from highest priority queue that has buffered packets
    - FCFS within priority class



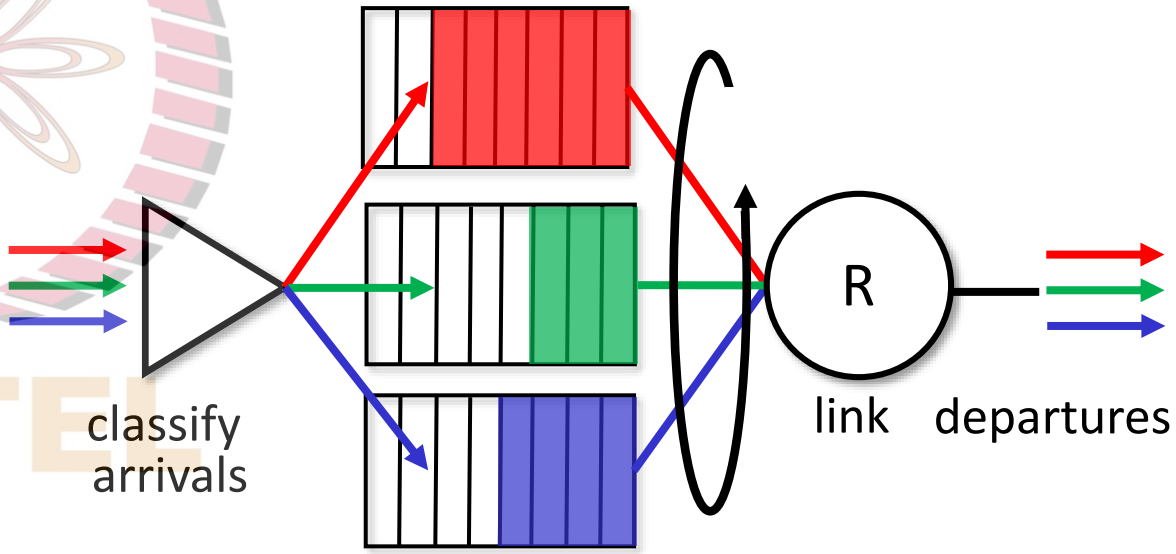


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Scheduling policies: round robin

Round Robin (RR) scheduling:

- arriving traffic classified, queued by class
  - any header fields can be used for classification
- server cyclically, repeatedly scans class queues, sending one complete packet from each class (if available) in turn



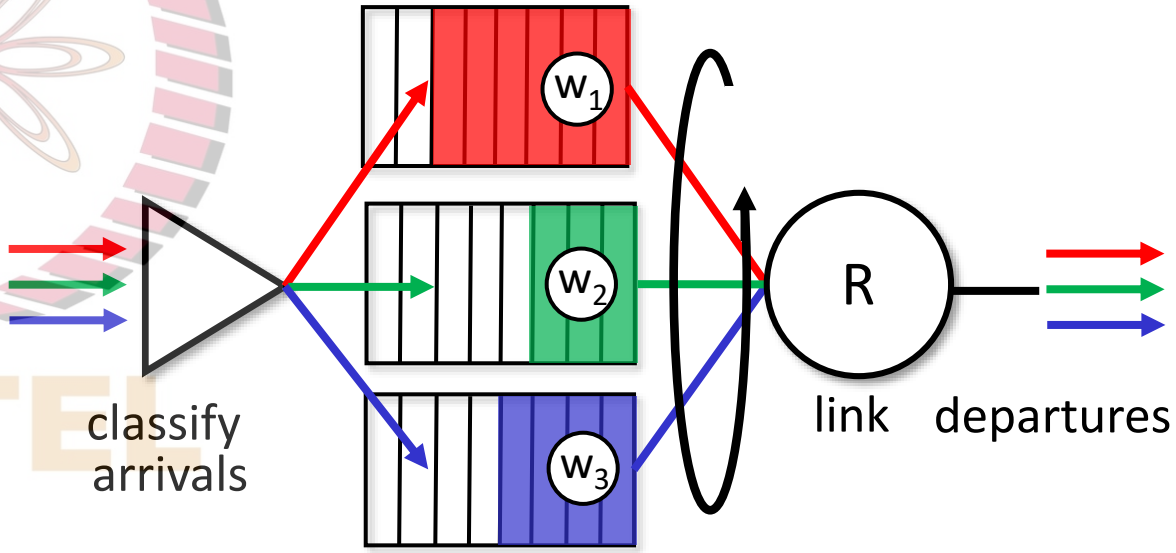
Scheduling policies: weighted fair queueing

Weighted Fair Queuing (WFQ):

- generalized Round Robin
- each class, *i*, has weight, w_i, and gets weighted amount of service in each cycle:

$$\frac{w_i}{\sum_j w_j}$$

- minimum bandwidth guarantee (per-traffic-class)



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Is Packet Scheduling worth the Effort ?

Network Neutrality

What is network neutrality?

- *Technical:* how an ISP should share/allocation its resources
  - Packet scheduling, buffer management are the *mechanisms*
- *Social, economic* principles
  - Protecting free speech
  - Encouraging innovation, competition
- Enforced *legal* rules and policies

Different states have different "takes" on network neutrality

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Sidebar: Network Neutrality

2015 US FCC *Order on Protecting and Promoting an Open Internet:* three "clear, bright line" rules:

- no blocking … "shall not block lawful content, applications, services, or non-harmful devices, subject to reasonable network management."
- no throttling … "shall not impair or degrade lawful Internet traffic on the basis of Internet content, application, or service, or use of a non-harmful device, subject to reasonable network management."
- no paid prioritization. … "shall not engage in paid prioritization."

2017: rollback of 2015 order
2024: rollback of 2017 rollback

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Network Neutrality in India

- Internet Service Providers (ISPs) must treat all data equally
- No blocking, throttling, paid prioritization

Laws

1. TRAI Regulation (2016)- "Prohibition of Discriminatory Tariffs for Data Services Regulations, 2016"

- Key rule: ISPs **cannot charge different prices for different websites/apps**
- Result: Banned **zero-rating services** like Facebook Free Basics

2. DoT (Department of Telecommunications) Rules (2018)

- This is the **most important step**: Government **accepted TRAI recommendations**. Net neutrality principles were **Added into telecom license agreements**